

Part A · Multiple Choice

2 points each

1. **d) richer in clay**

p. 174 — chemical weathering reactions, especially the formation of clay minerals, proceed fastest under warm conditions.

2. **d) Basalt**

p. 175 — quartz-rich materials such as granite and sandstone yield sandy soils instead. River-flood deposits also make good soils because they are rich in clay minerals.

3. **a) Less than 1%**

p. 182 — the mantle is about 68% and the core about 31%.

4. **d) Seafloor spreading**

p. 190 — hot mantle rises at a mid-ocean ridge, erupts as new seafloor, and pushes older crust outward, carrying the continents along like a conveyor belt.

5. **c) ocean-to-ocean convergent boundaries, as island arcs**

p. 195 — the older, denser oceanic plate subducts and a line of volcanoes grows on the upper plate. Most of the Ring of Fire is ocean-to-ocean convergence.

6. **d) reverse fault in which the fault plane is nearly horizontal**

p. 202 — rocks can slip many miles along thrust faults, which is how converging continental crust stacks up and rises into mountain ranges.

7. **c) 9.0**

p. 205 — the most powerful ever to strike Japan and among the top five known worldwide. Most of the damage came from the tsunami it generated rather than the shaking.

8. **a) Building a water reservoir on an active fault, where the water lubricates the fault and its weight stresses the bedrock**

p. 211 — the text also lists injecting sewage or chemical waste into a fault, and nuclear explosions, which is one way to detect that a nation has tested a bomb.

9. **c) About 10 times faster**

p. 214 — the ascending column may be kilometers to tens of kilometers across, spreading near the surface into a mushroom head tens to over 100 km wide.

10. **a) hot lava oozes into cold seawater and chills quickly on the outside**

p. 222 — the lava then behaves a little like toothpaste, forming blobs that pile around a seafloor vent. In terms of area there is very likely more pillow basalt on the seafloor than any other rock on Earth.

11. **c) Water**

p. 216 — followed typically by carbon dioxide, and then by sulphur dioxide.

12. **d) the wave's orbits are flattened and slowed by dragging on the bottom**

p. 229 — the wave begins to “feel the bottom” once the water is shallower than the wave base. The waves steepen, lean forward, and eventually break.

- 13. a) local change caused by uplift or subsidence of the crust**
p. 237 — usually from the gain or loss of glacial ice loading the crust. Several hundred meters of rebound has occurred around Hudson Bay since the last glaciation.
- 14. c) there is little or no input of clastic sediment from streams**
p. 235 — corals, algae, and shelled organisms need clear water. Upwelling currents help by bringing nutrients, so long as the water stays warmer than about 25°C.
- 15. d) sedimentary rocks have been folded or tilted and then eroded to varying degrees**
p. 247 — dendritic is the pattern for structureless rock, radial for volcanoes and domes, and deranged for basins with no coherent pattern.
- 16. a) more effective at keeping sediment in suspension**
p. 252 — fast streams tend to be turbulent, with chaotic flow paths and muddy water; slow streams tend toward laminar flow, straight-line paths, and clear water.
- 17. b) About 15%**
p. 252 — it includes ions such as calcium and chloride carried in solution, as opposed to the bedload and suspended load.
- 18. b) 4.5 m/sec**
p. 263 — it first rolls along in traction, then begins saltating as wind speed increases. Much of what we know comes from Brigadier Ralph Bagnold's desert experiments in the 1930s and 40s.
- 19. a) do not migrate at all, but lengthen over time**
p. 269 — they form where wind blows from more than one direction and sand is abundant. Seifs are the elongate variety with sharp knife-like crests.
- 20. b) wind comes from several different directions, and they grow vertically rather than laterally**
p. 269 — pyramidal mounds with slipfaces on three or more radiating arms, common to ergs. The Sand Hills of Nebraska are remnants of an ancient sand sea.
- 21. b) bajada**
p. 270 — Bajada is Spanish for inclination or descent. They are common in dry climates such as the Southwestern United States, where flash floods deposit alluvium over time.
- 22. b) outward from where the ice is thickest toward the thinner edges**
p. 280 — the areas they cover are generally flat, so the thickest central parts flow almost vertically downward while the peripheral parts flow outward. Alpine glaciers, by contrast, are controlled by the slope beneath the ice.
- 23. c) About 99%**
p. 280 — and about 68% of all of Earth's freshwater. The Antarctic Ice Sheet holds roughly 17 times as much ice as the Greenland Ice Sheet.
- 24. b) 80 meters**
p. 280 — enough to submerge most of Earth's major cities.
- 25. a) the base of the glacier is warm enough that a film of water forms between the ice and the material beneath**
p. 284 — if the base is cold the ice is stuck fast and all movement must occur by internal flow. Thin leading edges often freeze to their bed while the rest of the glacier keeps sliding, thrusting ice over the stuck portion.

Part B · Matching

8 points each

26. 1-c · 2-a · 3-f · 4-h · 5-g · 6-d · 7-e · 8-b

1 point each. *Hydrolysis* p. 170 · *Mass wasting* p. 167 · *Lithosphere* p. 183 · *Apparent polar wander* p. 189 · *Joint* p. 202 · *Elastic deformation* p. 200 · *Flux melting* p. 213 · *Tephra* p. 217.

27. 1-c · 2-f · 3-h · 4-g · 5-e · 6-a · 7-b · 8-d

1 point each. *Wave base* p. 229 · *Baymouth bar* p. 234 · *Point bar* p. 248 · *Porosity* p. 258 · *Ventifact* p. 263 · *Erg* p. 265 · *Firn* p. 281 · *Esker* p. 292.

Part C · Short Answer

3 points each

28. Soil is made largely of the products of weathering — clay minerals and quartz fragments — mixed with organic matter, so rock must be broken down first. Soil fails to develop on steep slopes, where erosion outpaces formation, and on newly created surfaces, since even ideal conditions require thousands of years.

Rubric: 1 pt — mechanical and chemical weathering supply the mineral fraction of soil (about 45%), dominated by clay minerals and quartz; without weathering there is nothing for organic matter to mix with. 1 pt each for two blockers, chosen from: steep slopes, where the rate of erosion exceeds the rate of soil formation; too little time, since soil takes thousands of years and recently glaciated or newly deposited surfaces are too young; too much water, which leaches nutrients and acidifies; too little water, which leaves salts and carbonates and little organic material; and areas of active mass wasting. (pp. 174–176)

29. P-waves slow at the mantle–core boundary, showing the outer core is less rigid than the mantle. S-waves stop there entirely, and since only solids resist a change in shape, the outer core must be liquid. S-waves reaching the inner core show it is solid.

Rubric: 1 pt — P-waves are fastest, travel by compression and expansion, and bend when passing between layers; they slow at the mantle–core boundary, producing the P-wave shadow zone between 104° and 140° from the focus. 1 pt — S-waves move material perpendicular to travel and can propagate only through solids, so their disappearance at that boundary proves the outer core is liquid. 1 pt — other lines of evidence agree: Earth's overall density exceeds that of crustal rock, so the core must be dense metal; Earth has a magnetic field, and iron and nickel are magnetic; and metallic meteorites resemble core material. (pp. 180–184)

30. At a subduction zone an enormous area of one plate is forced beneath another, generating reverse and thrust faults. Because magnitude depends on the area of the rupture surface and the amount of displacement, the vast contact area between the plates allows far larger ruptures than any other setting.

Rubric: 1 pt — earthquakes at convergent boundaries mark the motion of the subducting lithosphere plunging through the mantle, creating reverse and thrust faults. 1 pt — magnitude is proportional to the area of the rupture surface and the amount of displacement, and a subduction interface offers an enormous area. 1 pt — the largest ones occur in the upper section where the rock is relatively cool and therefore brittle enough to store stress before rupturing; quakes there can occur at well over 100 km depth, versus under 30 km at transform boundaries. Credit real examples: the 9.0 Tōhoku (2011), the Cascadia rupture of 1700, or the 9.2 Great Alaska (1964). (pp. 205–206, 211)

- 31. Erosional coasts are irregular with high relief and are worn into cliffs, caves, arches, and stacks; depositional coasts are relatively straight and low, building beaches, spits, and barrier islands. The main determining factor is the coast's history of tectonic activity.**

Rubric: 1 pt erosional — coasts uplifted within the past several million years have irregular shapes; refraction focuses wave energy on headlands, cutting sea caves, arches, stacks, cliffs, and wave-cut platforms, and straightening the coast over millions of years. Example: the Pacific coast of the U.S. and Canada. 1 pt depositional — coasts not uplifted for tens of millions of years are relatively straight with little relief, and build beaches, spits, baymouth bars, tombolos, and barrier islands from river-supplied sediment. Example: the U.S. Atlantic and Gulf coasts. 1 pt — the main determining factor is the history of tectonic activity, and both processes operate on almost every coast to some degree. (pp. 232–233)

- 32. Sand creeps and saltates up the gentle windward (stoss) slope to the crest, then spills down the steep leeward slip face and comes to rest at its angle of repose in the zone of stagnant air. Continual removal from the stoss side and deposition on the slip face moves the whole dune downwind.**

Rubric: 1 pt — the dune is asymmetric, with a gentle windward or stoss slope and a steep leeward slip face. 1 pt — saltating and creeping grains climb the stoss slope to the crest and pour over the top. 1 pt — they come to rest at the angle of repose, the maximum angle the sand pile will support, in the zone of stagnant air behind the crest; because sand is continually eroded from the stoss side and deposited on the slip face, the dune is unstable and migrates downwind, burying its own earlier layers and creating cross-beds. (pp. 264–266)

- 33. Continental glaciation erodes the land into relatively flat plains, with most relief coming from deposits such as drumlins, eskers, and moraines. Alpine glaciation produces dramatic topographic variety — U-shaped valleys, arêtes, cols, horns, cirques, and hanging valleys.**

Rubric: 1.5 pts continental — tends to erode the land surface into relatively flat bedrock plains, especially where the underlying rock is uniform in strength, eroding weaker rock preferentially where it is not; much of central and eastern Canada under the Laurentide Ice Sheet was planed this way, and the existing relief is largely depositional (drumlins, eskers, moraines). 1.5 pts alpine — produces great topographic variability, with the U-shaped valley as the key feature, plus arêtes, cols, horns, cirques, hanging valleys, and truncated spurs. Credit the note that both types produce drumlins, roches moutonnées, and glacial striations. (pp. 285–287)

34. See rubric.

Rubric: 2 pts — running water. Capacity is governed by flow velocity: the faster the flow, the larger the particles it carries. Material moves by traction, saltation, suspension, and solution, and turbulent flow suspends far more than laminar flow. Limits: it needs a gradient, and velocity collapses when a stream spreads onto its floodplain. Landforms: V-shaped valleys, canyons, meanders, oxbow lakes, point bars, cut banks, natural levées, floodplains, deltas, alluvial fans, and river terraces.

2 pts — wind. Needs fine, dry, loose material and, critically, an absence of vegetation, since plants both drag on the wind and bind particles down. Sand moves above about 4.5 m/sec, by traction (20–25%) and saltation (over 50%). Limits: it can lift only small particles, abrasion reaches barely a meter or two above the ground, and clay cohesion resists erosion. Landforms: desert pavement, blowout depressions, ventifacts, yardangs, and the whole family of dunes — barchan, transverse, parabolic, longitudinal, star.

2 pts — ice. Moves under its own weight by internal plastic deformation below 50–100 m and, where the base is warm, by basal sliding. The ice itself is soft (Mohs 1.5); the erosion is done by embedded rock fragments acting like sandpaper. Limits: it requires a cold climate and long timescales, and a glacier frozen to its bed cannot slide. Landforms: U-shaped valleys, arêtes, horns, cirques, hanging valleys, drumlins, striations, moraines, eskers, kettle lakes, and outwash plains.

2 pts — the argument. Any defensible position, well supported. The strongest case is running water: the text opens Unit 17 by calling streams “the most important agents of erosion and transportation of sediments on Earth’s surface,” responsible for much of the topography around us, and water operates across nearly every climate. A student may instead argue for ice on the grounds that glaciers covered at least 30% of land during the Pleistocene and left the present form of whole continents, or for wind on the grounds that deserts cover a large fraction of every continent. Credit the quality of the reasoning, not the choice.

(pp. 243, 251–253, 261–269, 280–292)

35. See rubric.

Rubric: 2 pts — earthquakes. Nearly 95% occur at plate boundaries, because that is where lithosphere is deformed until it ruptures (elastic rebound). About 80% ring the Pacific, which is lined with convergent and transform boundaries, and 15% fall in the Mediterranean-Asiatic Belt. Depth and size vary by boundary type: shallow and small at mid-ocean ridges where crust is young, thin, and hot; shallow but deadly at transform boundaries; deepest and largest at subduction zones.

2 pts — volcanoes. Magma forms in three tectonic settings by three distinct mechanisms: decompression melting at divergent boundaries, flux melting at convergent boundaries (water driven out of the subducting plate lowers the mantle's melting point), and decompression melting in mantle plumes. Composite volcanoes sit at subduction zones; shield volcanoes above mantle plumes; and most of Earth's volcanism occurs unseen along the spreading ridges.

2 pts — mountain ranges. Convergence builds mountains. Ocean-to-continent convergence produces continental volcanic arcs and batholiths; continent-to-continent convergence produces folding and thrust faulting with no volcanism. Tension at divergent settings breaks crust into basin-and-range topography along normal faults.

1.5 pts — coastlines. A coast's tectonic history determines whether it is erosional or depositional. Recently uplifted coasts are irregular and erosion-dominated, cut into cliffs, arches, and stacks; long-quiet coasts are straight, low, and depositional, building barrier islands. Tectonic sea-level change and post-glacial isostatic rebound shift the land relative to the sea.

0.5 pt — examples. At least three, drawn from: the San Andreas Fault (transform); the Cascades and Juan de Fuca subduction; the Himalayas (continent–continent); the Mid-Atlantic Ridge and the Great Rift Valley (divergent); Hawaii (mantle plume); the Ring of Fire; the Basin and Range; the Pacific versus Atlantic coasts of North America.

(pp. 192–197, 200–206, 213–214, 232–238)